

Section 1.4 Exercises

Directions: For exercises 1-5, use the given functions to find a. $(f + g)(x)$ b. $(f - g)(x)$ c. $(f \cdot g)(x)$ d. $\left(\frac{f}{g}\right)(x)$

1. $f(x) = x^2 + 2x$ and $g(x) = 6 - x^2$

2. $f(x) = -3x^2 + x$ and $g(x) = 5$

3. $f(x) = 2x^2 + 4x$ and $g(x) = \frac{1}{2x}$

4. $f(x) = \frac{1}{x-4}$ and $g(x) = \frac{1}{6-x}$

5. $f(x) = 3x^2$ and $g(x) = \sqrt{x-5}$

6. Given $f(x) = 2x^2 + 1$ and $g(x) = 3x - 5$, find the following:

a. $f(g(2))$

b. $f(g(x))$

c. $g(f(x))$

d. $(g \circ g)(x)$

e. $(f \circ f)(-2)$

Directions: For exercises 7-12, use the given functions to find $f(g(x))$ and $g(f(x))$. Simplify your answers.

7. $f(x) = x^2 + 1$, $g(x) = \sqrt{x+2}$

8. $f(x) = \sqrt{x} + 2$, $g(x) = x^2 + 3$

9. $f(x) = |x|$, $g(x) = 5x + 1$

10. $f(x) = \sqrt[3]{x}$, $g(x) = \frac{x+1}{x^3}$

11. $f(x) = \frac{1}{x-6}$, $g(x) = \frac{7}{x} + 6$

12. $f(x) = \frac{1}{x-4}$, $g(x) = \frac{2}{x} + 4$

Directions: For exercises 13-14, use each set of functions to find $f(g(h(x)))$. Simplify your answers.

13. $f(x) = x^4 + 6$, $g(x) = x - 6$, and $h(x) = \sqrt{x}$

14. $f(x) = x^2 + 1$, $g(x) = \frac{1}{x}$, and $h(x) = x + 3$

15. Given $f(x) = \frac{1}{x}$ and $g(x) = x - 3$, find the following:

a. $(f \circ g)(x)$

b. $(g \circ f)(x)$

c. $\left(\frac{f}{g}\right)x$

16. Given $f(x) = \sqrt{2 - 4x}$ and $g(x) = -\frac{3}{x}$, find $(g \circ f)(x)$.

17. Given the functions $f(x) = \frac{1-x}{x}$ and $g(x) = \frac{1}{1+x^2}$, find the following:

a. $(g \circ f)(x)$

b. $(g \circ f)(2)$

Directions: For exercises 18-33, find functions $f(x)$ and $g(x)$ so the given function can be expressed as $h(x) = f(g(x))$.

18. $h(x) = (x + 2)^2$

19. $h(x) = (x - 5)^3$

20. $h(x) = \frac{3}{x-5}$

21. $h(x) = \frac{4}{(x+2)^2}$

22. $h(x) = 4 + \sqrt[3]{x}$

23. $h(x) = \sqrt[3]{\frac{1}{2x-3}}$

24. $h(x) = \frac{1}{(3x^2-4)^{-3}}$

25. $h(x) = \sqrt[4]{\frac{3x-2}{x+5}}$

26. $h(x) = \left(\frac{8+x^3}{8-x^3}\right)^4$

27. $h(x) = \sqrt{2x+6}$

28. $h(x) = (5x - 1)^3$

29. $h(x) = \sqrt[3]{x-1}$

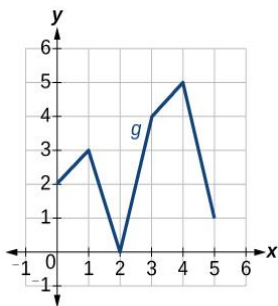
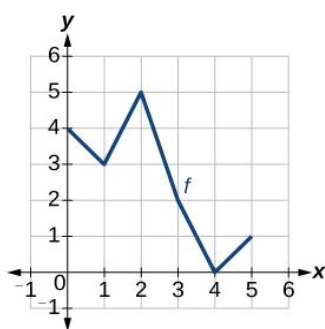
30. $h(x) = |x^2 + 7|$

31. $h(x) = \frac{1}{(x-2)^3}$

32. $h(x) = \left(\frac{1}{2x-3}\right)^2$

33. $h(x) = \sqrt{\frac{2x-1}{3x+4}}$

Directions: For exercises 34-41, use the graphs of f and g to evaluate the expressions.



34. $f(g(3))$

35. $f(g(1))$

36. $g(f(1))$

37. $g(f(0))$

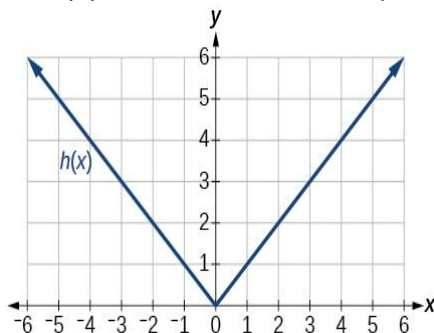
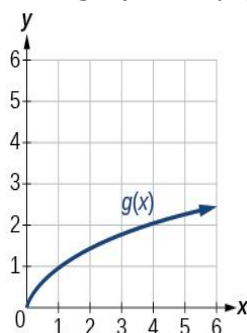
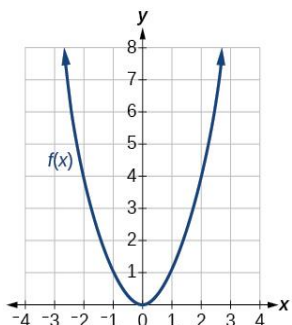
38. $f(f(5))$

39. $f(f(4))$

40. $g(g(2))$

41. $g(g(0))$

Directions: For exercises 42-49, use graphs of $f(x)$, $g(x)$, and $h(x)$, to evaluate the expressions.



42. $g(f(1))$

43. $g(f(2))$

44. $f(g(4))$

45. $f(g(1))$

46. $f(h(2))$

47. $h(f(2))$

48. $f(g(h(4)))$

49. $f(g(f(-2)))$

Directions: For exercises 50-57, use the function values for f and g shown in the table below to evaluate each expression.

x	$f(x)$	$g(x)$
0	7	9
1	6	5
2	5	6
3	8	2
4	4	1
5	0	8
6	2	7
7	1	3
8	9	4
9	3	0

50. $f(g(8))$

51. $f(g(5))$

52. $g(f(5))$

53. $g(f(3))$

54. $f(f(4))$

55. $f(f(1))$

56. $g(g(2))$

57. $g(g(6))$

Directions: For exercises 58-63, use the function values for f and g shown in the table below to evaluate each expression.

x	$f(x)$	$g(x)$
-3	11	-8
-2	9	-3
-1	7	0
0	5	1
1	3	0
2	1	-3
3	-1	-8

58. $(f \circ g)(1)$

59. $(f \circ g)(2)$

60. $(g \circ f)(2)$

61. $(g \circ f)(3)$

62. $(g \circ g)(1)$

63. $(f \circ f)(3)$

Directions: For exercises 64-67, use each pair of functions to find $f(g(0))$ and $g(f(0))$.

64. $f(x) = 4x + 8$, $g(x) = 7 - x^2$

65. $f(x) = 5x + 7$, $g(x) = 4 - 2x^2$

66. $f(x) = \sqrt{x+4}$, $g(x) = 12 - x^3$

67. $f(x) = \frac{1}{x+2}$, $g(x) = 4x + 3$

Directions: For exercises 68-71, use the functions $f(x) = 2x^2 + 1$ and $g(x) = 3x + 5$ to evaluate or find the composite function as indicated.

68. $f(g(2))$

69. $f(g(x))$

70. $g(f(-3))$

71. $(g \circ g)(x)$